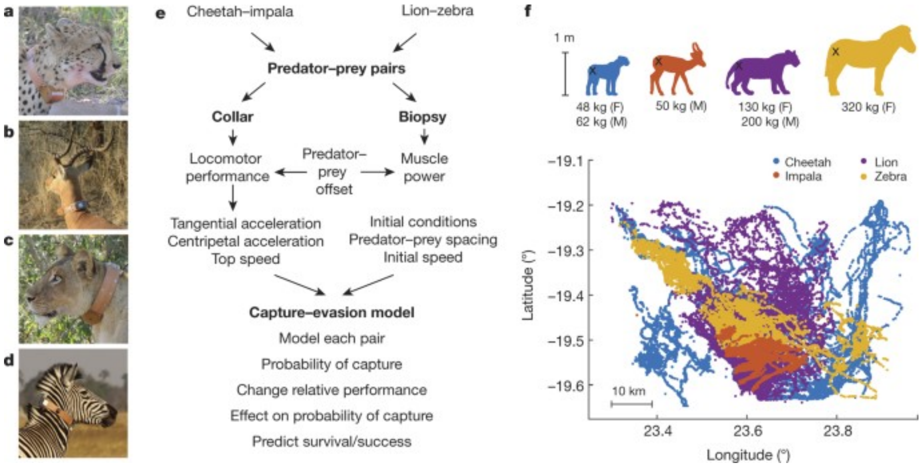
Lab 2: Environmental Simulators


a
b
c
d
e
Cheetah–impala
Lion–zebra
Predator–prey pairs
Collar
Biopsy
Locomotor performance
Predator–prey offset
Muscle power
Tangential acceleration
Centripetal acceleration
Top speed
Initial conditions
Predator–prey spacing
Initial speed
Capture–evasion model
Model each pair
Probability of capture
Change relative performance
Effect on probability of capture
Predict survival/success
f
1 m
48 kg (F)
62 kg (M)
50 kg (M)
130 kg (F)
200 kg (M)
320 kg (F)
Cheetah
Impala
Lion
Zebra
−19.1
−19.2
−19.3
−19.4
−19.5
−19.6
Latitude (°)
10 km
23.4
23.6
23.8
Longitude (°)

1 Introduction

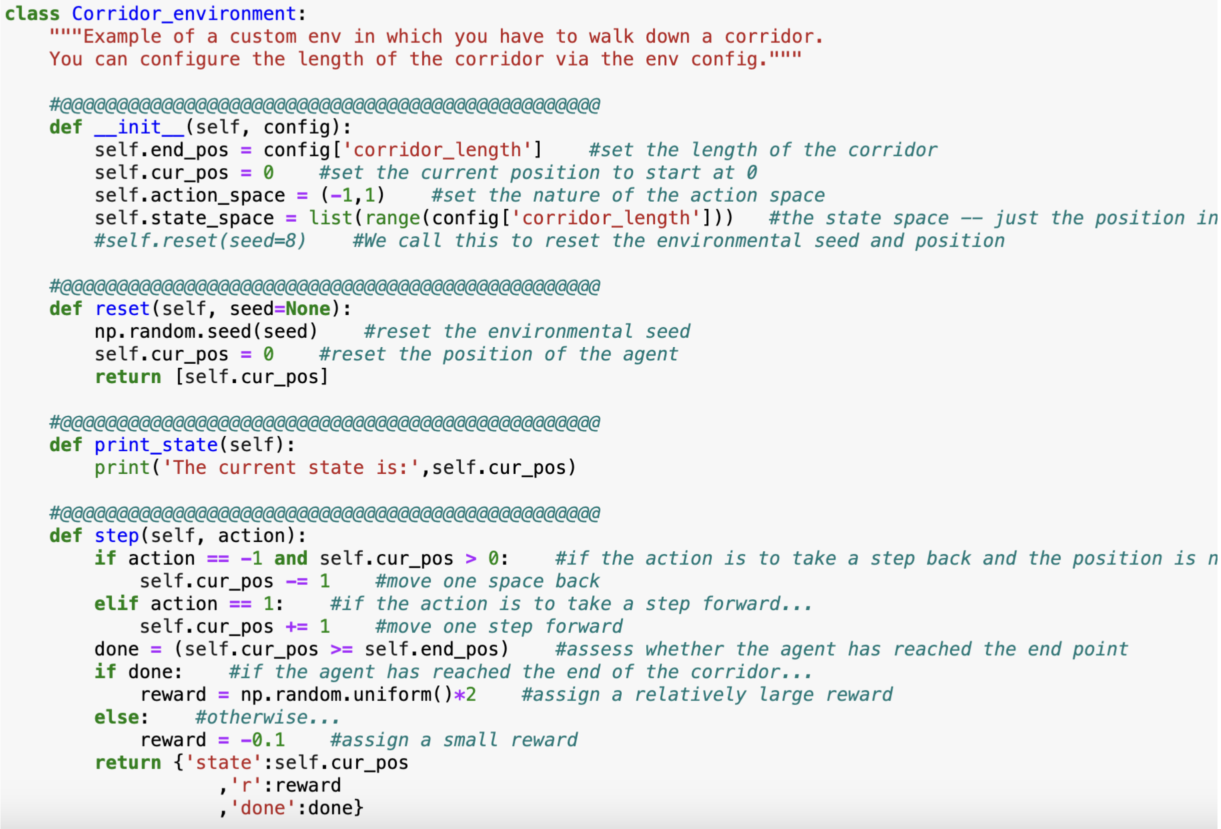
Environmental simulators play a crucial role in RL by providing a safe and controlled environment where agents can learn and explore without real-world consequences. These simulators allow RL algorithms to iterate through vast numbers of trials, enabling the agent to practice and improve its decision-making process through trial and error. In a simulator, we can model complex environments, introduce variability, and manipulate conditions to better understand how an agent learns. The ability to run experiments at scale and speed that would be impractical in the real world makes simulators indispensable for developing and fine-tuning RL algorithms, especially when real-world testing is costly, dangerous, or slow. In this lab we will practice implementing a relatively simple environment and simulating an agent making random decisions.

1.1 The Gymnasium Standard

Gymnasium, AKA OpenAI Gym, is a popular toolkit used in RL for developing and comparing RL algorithms. Gymnasium provides a relatively simple, standardized interface to a wide range of environments, from simple tasks like balancing a pole on a cart to more complex video games and robotics simulations. Many RL packages are structured to work with Gymnasium. For this reason, we will aim to build environments using a gymnasium-like structure.

In general, a gymnasium environment is specified as a python class that records the current state, specifies the state and action space structure, and has at least two functions: a **reset** function and a **step** function.

- In RL there are two kinds of environments: 1) continuing environments that essentially run forever without stopping and 2) episodic environments that begin, run for a



```
class Corridor_environment:
    """Example of a custom env in which you have to walk down a corridor.
    You can configure the length of the corridor via the env config."""

    #@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
    def __init__(self, config):
        self.end_pos = config['corridor_length']    #set the length of the corridor
        self.cur_pos = 0    #set the current position to start at 0
        self.action_space = (-1,1)    #set the nature of the action space
        self.state_space = list(range(config['corridor_length']))   #the state space -- just the position in
        #self.reset(seed=8)    #We call this to reset the environmental seed and position

    #@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
    def reset(self, seed=None):
        np.random.seed(seed)    #reset the environmental seed
        self.cur_pos = 0    #reset the position of the agent
        return [self.cur_pos]

    #@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
    def print_state(self):
        print('The current state is:',self.cur_pos)

    #@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
    def step(self, action):
        if action == -1 and self.cur_pos > 0:    #if the action is to take a step back and the position is n
            self.cur_pos -= 1    #move one space back
        elif action == 1:    #if the action is to take a step forward...
            self.cur_pos += 1    #move one step forward
        done = (self.cur_pos >= self.end_pos)    #assess whether the agent has reached the end point
        if done:    #if the agent has reached the end of the corridor...
            reward = np.random.uniform()*2    #assign a relatively large reward
        else:    #otherwise...
            reward = -0.1    #assign a small reward
        return {'state':self.cur_pos
                ,'r':reward
                ,'done':done}
```

Figure 1: An example of a custom environment with a gymnasium-like structure.

finite number of turns, and then stop. The **reset** function is used to return episodic environments to a starting state so that a new episode may begin.

- The **step** function is typically used to evolve the environment. It takes in an agent action and returns an update to the current state, a generated reward, and an indicator for whether or not the episode has ended.

2 Install and Explore Gymnasium

To begin, install Gymnasium and the ALE Atari integration. The following commands entered at the command line enabled me to set up a virtual environment on Rosie in which I could run the environment and train an agent, with a kernel recognized by jupyter notebooks:

1. conda create -n py_Gym python=3.10
2. conda activate py_Gym
3. pip install ale-py
4. pip install autorom[accept-rom-license]

5. AutoROM - -accept-license
6. pip install “gymnasium[atari,accept-rom-license]” - -force-reinstall
7. pip install torch
8. conda install matplotlib
9. conda install ipykernel
10. python -m ipykernel install –user –name=py_Gym - -display-name=“Python py_Gym”

Gymnasium contains a large number of pre-built environments spanning a variety of different state, action, and reward structures. Basic documentation is available here.

1. Choose any prebuilt gymnasium environment.
2. Build the environment in your notebook using the make() method.
3. Write code to execute a simple loop in which you supply the environment an action and the environment returns to reward and updated state.
4. Simulate 1000 episodes with randomly generated agent actions.

3 Implement two custom environments

Next take the following environments and implement them in python using a gymnasium-like structure.

Environment 1 – a racetrack. Consider driving a race car around a turn. You want to go as fast as possible, but not so fast as to run off the track. The racetrack is a grid – it begins with a vertical section 10 cells wide and 30 cells high, followed by a right turn into a horizontal section 15 cells wide and 10 cells tall. The starting line is the row of cells at the bottom of the first section. The finish line is the column of cells at the right of the horizontal section. The car begins at any cell on the starting line and each turn it is at one of the grid positions. The velocity is also discrete, a number of grid cells moved horizontally and vertically per time step. The actions are increments to the velocity components. Each may be changed by $+1$, -1, or 0 in each step, for a total of nine (3×3) actions. The vertical velocity component is restricted to be nonnegative. Both velocity components are restricted to be less than 5, and they cannot both be zero except at the starting line. Each episode begins in one of the randomly selected start states with both velocity components zero and ends when the car crosses the finish line. If the car attempts to go through the track boundary (anywhere by the finish line) it crashes and the episode ends.

Environment 2 – a chase. Consider a chase on a square field with each side of length n. In the field is a predator and a player. The player begins in cell position $(0, 0)$. The predator begins in a randomly generated cell position. At position (n, n) is the player’s base.

Unlike the racetrack example, the player and predator can be at any location in the field, i.e. any real number between 0 and n. The velocity of the player is continuous, a distance moved horizontally and vertically per time step. The actions are increments to the velocity components. Each may be changed by $+1$, -1, or 0 in each step, for a total of nine (3×3) actions. In this case they are stochastic; given a horizontal velocity at time t of $V_H^{(t)}$:

$$V_H^{(t+1)} = \begin{cases} V_H^{(t)} + U([0.5, 1.5]) & \text{if } a_H^{(t)} = 1 \\ V_H^{(t)} & \text{if } a_H^{(t)} = 0 \\ V_H^{(t)} + U([-0.5, -1.5]) & \text{if } a_H^{(t)} = -1. \end{cases}$$

The vertical velocity is the same. Both velocity components are restricted to be no more than 5. Each turn, randomly choose either the predator or the player to move first. The predator moves a distance of no more than 4 directly toward the player. The player moves according to their current velocity (but must stay within the field of play). If the predator lands on the cell occupied by the player then the player is caught and the episode ends. If the player is within a distance of 0.5 of their base without being caught then they escape and the episode ends.

1. Implement these two environments using gymnasium-like python classes. Make sure to include all the relevant elements of said class: the ability to track the current state; the **reset** function; and the **step** function.
2. Simulate 1000 episodes with randomly generated agent actions.

4 Make Learning Tough!

If you do reinforcement learning enough you'll probably come to find that the information and reward structure of the environment is at least as important as the algorithms you use to train an agent in determining how effect the agent will be. Here you will consider what might make learning difficult for an agent.

1. Choose one of your custom environments (the racetrack or the chase) and modify it so that it becomes significantly harder for an RL agent to solve. Possible modifications include deceptive rewards, partial observability, stochastic transitions, some sort of nonstationarity, randomly generated obstacles, etc.
2. Simulate 1000 episodes with randomly generated agent actions.

Lab Assignment Report

Write up your analysis in a separate report of no more than 2 pages. Your report should answer the following questions:

1. For the gymnasium environment describe the task the agent must do. What does success mean? Report an appropriate summary of the state across the 1000 episodes. Also, report the average reward across episodes at each time step. How often does it succeed at its task with random actions?

2. For each custom environment, plot a heatmap of the agent's position. How often does it succeed at its task with random actions (crossing the finish line for the track, reaching base for the chase)?
3. Describe how you would seek to reward the agent in each custom environment. Can you identify any challenges the agent might face in learning given your suggested approach to rewards? How difficult was reward design?
4. Describe how you would conceptualize the state in each custom environment – what information do you think the agent would need to manage its task?
5. Compare the original and modified environments over your 1,000 episode runs. Compare success rate, average reward, and episode length, and summarize your findings. Do you think your modification would make it more difficult for an agent to learn? Why or why not?
6. How would the ideas in this lab apply to autonomous vehicles, robotics, or recommender systems?

5 Submission Instructions

- Write up the answers to the questions in a short word document; aim for no more than 2 pages of written text and include all graphics generated. Add footnotes identifying which sentence addresses which questions. Write in complete sentences organized into paragraphs – your goal is to explain what you've done and what you've learned to your audience (me!). Include the appropriate plots you've generated as mentioned above. Convert this to pdf and submit it. Submit your .ipynb file and an .html of it as well.
- Please use the following naming conventions:
  - Lab_2_[YOUR NAME].pdf
  - Lab_2_[YOUR NAME].html
  - Lab_2_[YOUR NAME].ipynb
- The grading rubric for this assignment will be available in Canvas.
- NO OTHER SUBMISSION TYPES WILL BE ACCEPTED.
- You are welcome to use AI tools for implementation as you code up your solution, but the emphasis is on experimental design, analysis, and scientific reasoning. If you do this you must include all prompts used as an appendix in your written report or provide a shareable link to them. Any uncited use of generative AI (i.e. prompts not provided) will be considered plagiarism. You may not use generative AI to write your report.